# MBR & FAT32 Analysis

**TOTAL SCORE: _____ / 63 PTS**

**Required Files:** `mbr.dd`, `fat32.dd` (**Download ZIP Here**)
**Tools:** FTK Imager, HxD, or any Hex Editor.

## Task 1: MBR Analysis

**GRADING POLICY: All or Nothing. Full points are awarded only for completely correct data.**

### Q1. Partition Table Completion

**16 Points Total**

Complete using values from `mbr.dd`. Reminder: LBA values are Little Endian.

| # | Bootable? (1pt) | Sys ID (1pt) | LBA Start (1pt) | LBA Size (1pt) |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |

### Q2. Disk Signature

**1 Point**

What hex value is found for the disk signature?

______________________

### Q3. Linux Swap Identification

**2 Points**

Which partition is Linux Swap? Calculate byte offset of start: ____________________

### Q4. Origin Suggestion

**2 Points**

What does the signature at `0x1B8-0x1BB` suggest about the disk origin?

## Task 2: FAT32 Analysis

**GRADING POLICY: All or Nothing. Full points are awarded only for completely correct data.**

### Q5. Volume Label

1 Point

Name found at byte offset `0x47`: ____________________

### Q6. BPB Field Recording

12 Points Total

| Field | Raw Hex (1pt) | Decimal (1pt) |
|---|---|---|
| Bytes per sector (0x0B) | | |
| Sectors per cluster (0x0D) | | |
| Number of reserved sectors (0x0E) | | |
| Number of FATs (0x10) | | |
| Sectors per FAT (0x24) | | |
| Root cluster number (0x2C) | | |

### Q7. Data Region Start

2 Points

*Data start = Reserved Sectors + (Number of FATs * Sectors per FAT)*

Resulting Sector: ____________________

### Q8. Data Region Byte Offset

1 Point

*Byte Offset = Sector number * Bytes per sector*

Resulting Offset: ____________________

### Q9. Cluster Size

1 Point

*Cluster size = Sectors per cluster * Bytes per sector*

Bytes: ____________________

### Q10. REPORT.TXT Directory Entry

7 Points

Full 32-byte Entry: ____________________________________________________________

| Field | Raw Hex/Value (1pt) |
|---|---|
| Filename (8 Bytes) | |
| Extension (3 Bytes) | |
| Attribute Byte (1 Byte) | |
| Starting Cluster (High) | |
| Starting Cluster (Low) | |
| Starting Cluster (Combined) | |
| File Size (4 Bytes) | |

## Q11. File Byte Offset

2 Points

*Byte Offset = (Data Start Sector + (Starting Cluster - 2) * Sectors per Cluster) * Bytes per Sector*

Offset: ___________________

## Q12. FAT Entry Offset

1 Point

FAT entry offset for starting cluster: ___________________

## Q13. FAT Chain Pointer

2 Points

Hex Value: ______________ | Points to Cluster (Dec): ______________

## Q14. FAT Chain Table

9 Points Total

| Cluster (1pt) | FAT Entry Offset (1pt) | Next Cluster (1pt) |
|---|---|---|
| | | |
| | | |
| | | |

## Q15. End Marker

1 Point

Value marking end of chain: ___________________

## Q16. Fragmentation Forensics

3 Points

Are clusters used by REPORT.TXT contiguous? Forensic significance: ___________________

### BONUS: Manual Recovery

5 Points

Recover the full contents of **REPORT.TXT** using the offsets calculated above.

### Grading Summary

Task 1 Total: ____ / 21 pts | Task 2 Total: ____ / 42 pts | Bonus Points: ____ / 5 pts

**Grand Total Score: ____ / 63 pts**

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**Created by Tyler Lofton**